

# THE LEVELING LINE

JING-ZHANG AI INNOVATION BELT

The street at eye level and the kit of parts

THE STREET AT EYE LEVEL AND THE KIT OF PARTS

jiangmuran · Concept advice; not a substitute for statutory planning

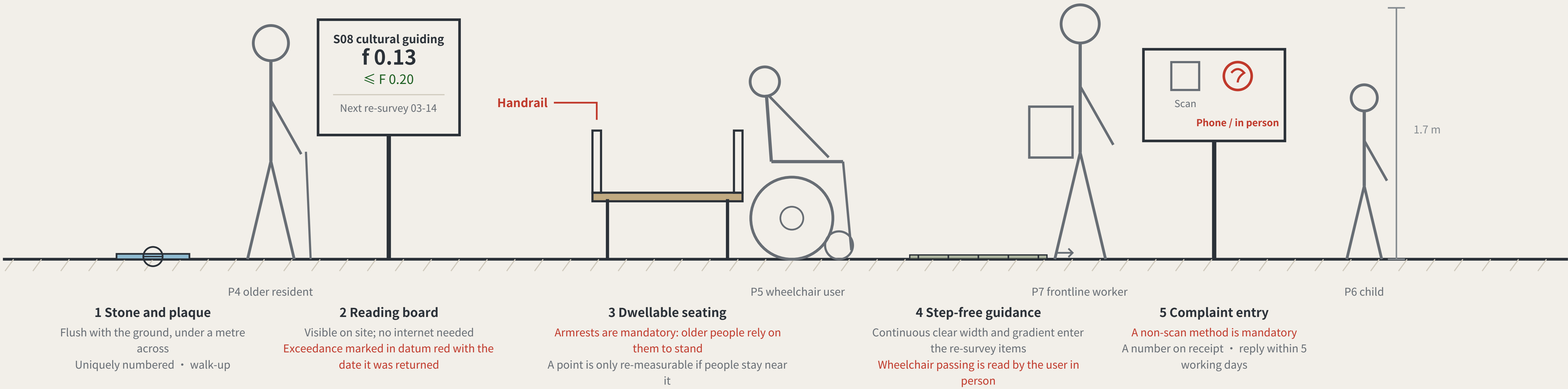
## THE BENCHMARK ON THE STREET AND HOW THE KERB IS SHARED

FIG.10

### What a benchmark looks like at eye level

A BENCHMARK AT EYE LEVEL · third-order BM-301 · elevation 1 m = 138 units

a type drawing; it carries no orientation



Two of the five standard components are hard constraints: seating must have armrests, and the complaint entry must offer a non-scan method. A rule you cannot see in the drawing is a rule the drawing does not contain.

### The same kerb, after snow clearance

In winter the order becomes one real trade-off

#### Piled on the device band

Devices stop; S06 falls back to staffed transfer

#### Piled on the pedestrian band

Pedestrians pushed into the carriageway - the worst of the three

#### Piled on the kerbside parking band

The one item in the order that may be lost first

This proposal takes this one

### Kerb allocation in priority order

KERB ALLOCATION IN PRIORITY ORDER · sequence and precedence only, no widths

#### 1 Emergency and fire access

Never occupied

#### 2 Step-free boarding, turning

Not occupied

#### 3 Walking and dwelling

Never squeezed below the level-of-service reserve

#### 4 Public transport and cycle parking

#### 5 Device charging and standby

Sited only from what remains after the four above

#### 6 Kerbside car parking

satisfied first →

→ allocated last

The order is itself a position: device charging sits behind pedestrians and accessibility, which means introducing devices must not be paid for with degraded walking conditions. Winter snow storage must be reserved at this stage alongside the device and pedestrian lanes — piled on the device lane the devices stop; piled on the pedestrian lane people are pushed toward the carriageway, which is worse.

The elevation is drawn at 1 m = 138 units. The figures are a scale reference and a statement about who is present, not a rendering. The section expresses sequence and precedence only and carries no widths — capacity must be computed on site from measured clear width, and a figure not derived from measurement is fabricated certainty. Component specifications are in geometry/public\_space.geojson and the scenario cards; positions await official boundaries and title verification.

THE LEVELING LINE · Haidian, Beijing · jiangmuran · Concept advice on provisional boundaries; not a substitute for statutory planning Data to 2026-08-17 (sources.json; 1 not obtained)

Plan EPSG:4548/4326 · heights relative; the 1985 national datum is not used

## LANDMARKS, KIT OF PARTS, SIGNAGE SYNTAX AND OPERATING CYCLE

FIG.09

a type drawing; it carries no orientation

### Three AI pilgrimage landmarks

THREE LANDMARKS · spoken in engineering language; no spectacle

#### L1 The origin benchmark stone

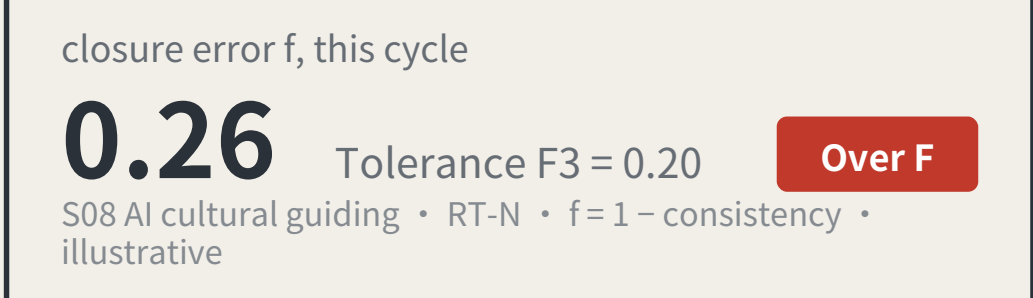
BM-0 · AI Origin Community



Every merged proposal's number is set into the paving — contributors' GitHub IDs are inscribed here. It is not grand. It is accurate, and must be accurate enough to be reused a century later.

#### L2 The closure stele

BM-1 · Zhongzhiyuan



A public reading wall, updated live, showing each scenario's closure error against its tolerance. Exceedances are marked in datum red. The value of this landmark is that it is allowed to look bad.

#### L3 return-to-datum point

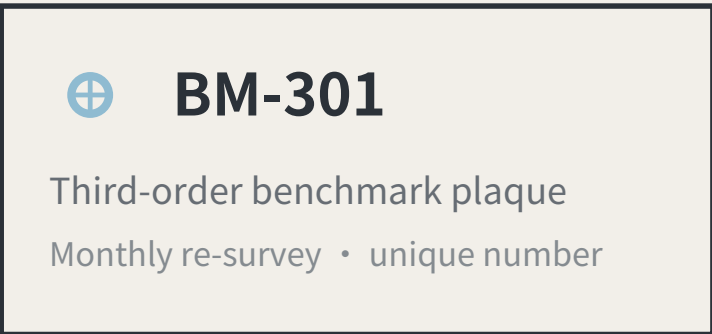
BM-2 · Dazhongsi



An annual civic ceremony space: the line's readings are zeroed here, tolerance revisions are read out, and scenarios returned for re-survey in the past year have their disposition explained. On ordinary days it is a public dwelling space.

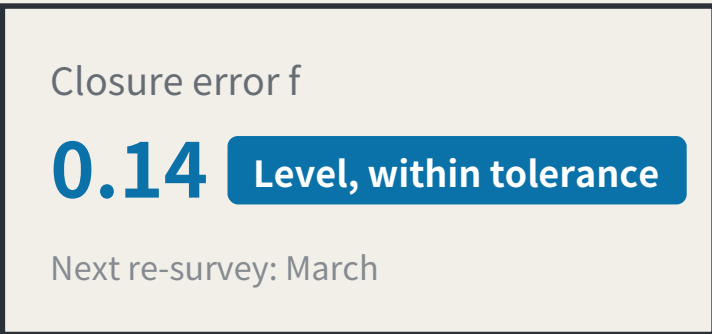
### Kit of parts for public space: five standard components

KIT OF PARTS · common specifications and open drawings, so any new node joins in one language



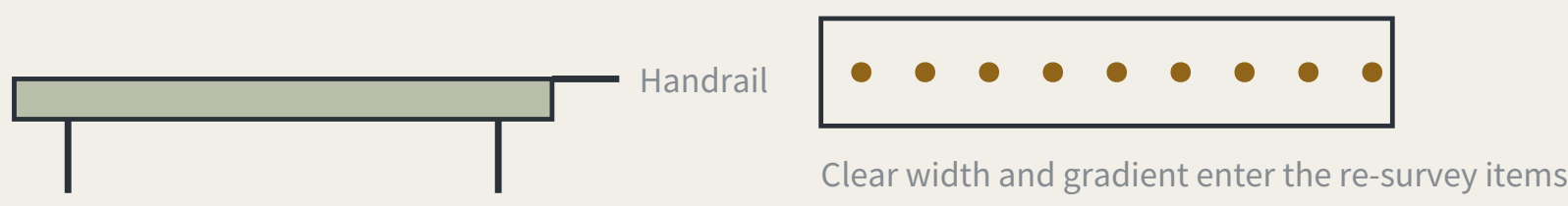
#### 1 Stone and plaque

The number is position, order and cycle



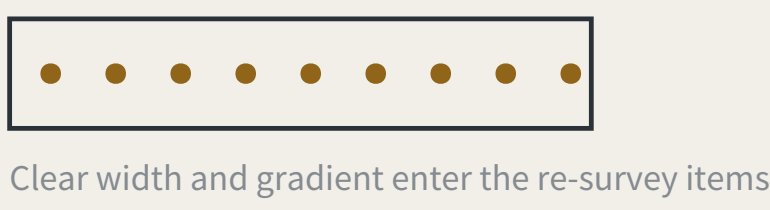
#### 2 Reading board

Visible on site; no internet needed



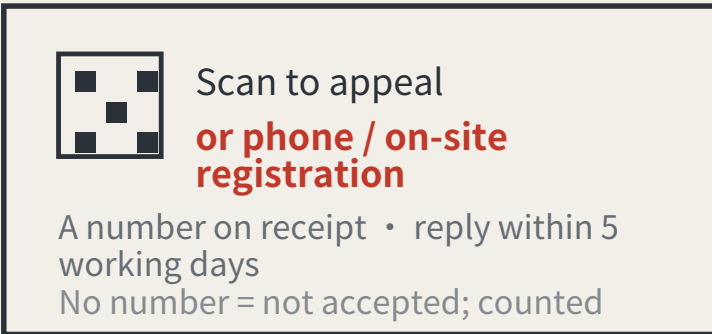
#### 3 Dwellable seating

With armrests: older people rely on them to stand



#### 4 Step-free guidance

Read by the users themselves



#### 5 Complaint entry

A non-scan method is mandatory, or the right does not exist for P4

### Signage numbering syntax (agent.5)

A visitor reading the number knows which order they are standing at and how often it is re-measured

|             |                       |                                         |
|-------------|-----------------------|-----------------------------------------|
| BM-0        | Origin benchmark      | Annual origin and closure computation   |
| BM-1 / BM-2 | First-order benchmark | Annual re-survey                        |
| BM-2x       | 2nd-order point       | Quarterly re-survey                     |
| BM-3xx      | 3rd-order point       | Monthly re-survey                       |
| RT-N / RT-S | Closing route         | Semi-annual re-survey of the whole line |

### Operations are organised by re-survey cycle, not by festival calendar (agent.6)

Events are therefore governance actions, not publicity

|             |                   |                                  |                                 |
|-------------|-------------------|----------------------------------|---------------------------------|
| Monthly     | Community day     | third order BM-301/BM-302/BM-303 | P4 · P5 · P7                    |
| Quarterly   | Scenario open day | second order BM-21/BM-22         | P2 · P3                         |
| Semi-annual | Route re-survey   | first order, routes RT-N / RT-S  | P1 · P2                         |
| Annual      | Zeroing ceremony  | L3 return-to-datum point         | all four review parties present |

Developer conversion path: take part in re-survey → propose a scenario → enter the test field → obtain closure clearance → operate  
International communication uses published readings as its material, never commitments.

The f=0.26 / f=0.14 on the reading boards are illustrative, not measured results. f is always a deviation: larger is worse, and f ≤ F is the passing test. Landmark siting requires heritage and engineering review; this proposal does not pre-empt it. The components are a directional proposal for a professional team to develop and are not construction drawings.

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THE LEVELING LINE

JING-ZHANG AI INNOVATION BELT

From region to site: coordination interfaces and the key areas

FROM REGION TO SITE

jiangmuran • Concept advice; not a substitute for statutory planning

REGIONAL COORDINATION INTERFACES: EXTENDING THE LEVELLING NETWORK

FIG.11

An executable form of coordination: how two levelling networks join

AN EXECUTABLE FORM OF COORDINATION: HOW TWO LEVELLING NETWORKS JOIN



A joint observation produces a number, not an intention:

Each network reads the same public question set; the difference is the cross-network closure error. Within a jointly agreed tolerance, the two networks' records may be mutually recognised; over it, each re-surveys, and no declaration of partnership substitutes for a reading.

Nobody gives up authority: a joint observation asks only that conventions be comparable, not that organisations merge.

What each of the five partners exchanges, and what it does not

| Partner               | Stage                                            | Exchanges                                                                                       | Explicitly does not exchange                                                                                               | Precondition for the interface                                      |
|-----------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
| BDA (Yizhuang)        | Volume production and high-level AV road testing | Mutual recognition of closure records across complementary speed domains                        | Not one admission: an F1 pass here admits a device to pedestrian density, not to a carriageway, and the reverse also holds | Both parties confirm the speed-domain split and test protocols      |
| Future Science City   | Corporate research institutes and engineering    | Scenario demand and real user density                                                           | No claim on IP; no scheduling on anyone's behalf                                                                           | Wing benchmarks built and a first baseline taken                    |
| Huairou Science City  | Large facilities and basic research              | The measurement method itself: how f is defined, cycle length, the statistics behind revising F | No scenario data is exchanged; the coordination is methodological, not applied                                             | The closure convention published and open to methodological review  |
| Beiwei Community      | The innovation community the taskbook names      | Reciprocal joint points; exchanged review parties                                               | This proposal does not know its positioning and does not set its benchmark order                                           | Each party names at least one publicly reachable joint point        |
| Beijing-Tianjin-Hebei | Cross-provincial coordination                    | Cross-provincial recognition of the tolerance convention                                        | No direct recognition of records; without a shared convention they are not comparable                                      | Tolerance definition and revision procedure agreed across provinces |

None of the five rows has been confirmed by the party named.

No internal plans are known, nobody is committed, no agreement is assumed; each row is an interface to evaluate independently.

Each partner's stage is described from publicly available general positioning and is unconfirmed by them. The speed-domain relationship in the first row must be settled against both parties' actual test protocols. This sheet proposes interfaces; it commits nobody and is no government determination.

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Plan EPSG:4548/4326 • heights relative; the 1985 national datum is not used

THE THREE KEY AREAS, THE TWO WINGS AND THE BENCHMARK LAYOUT

FIG.04

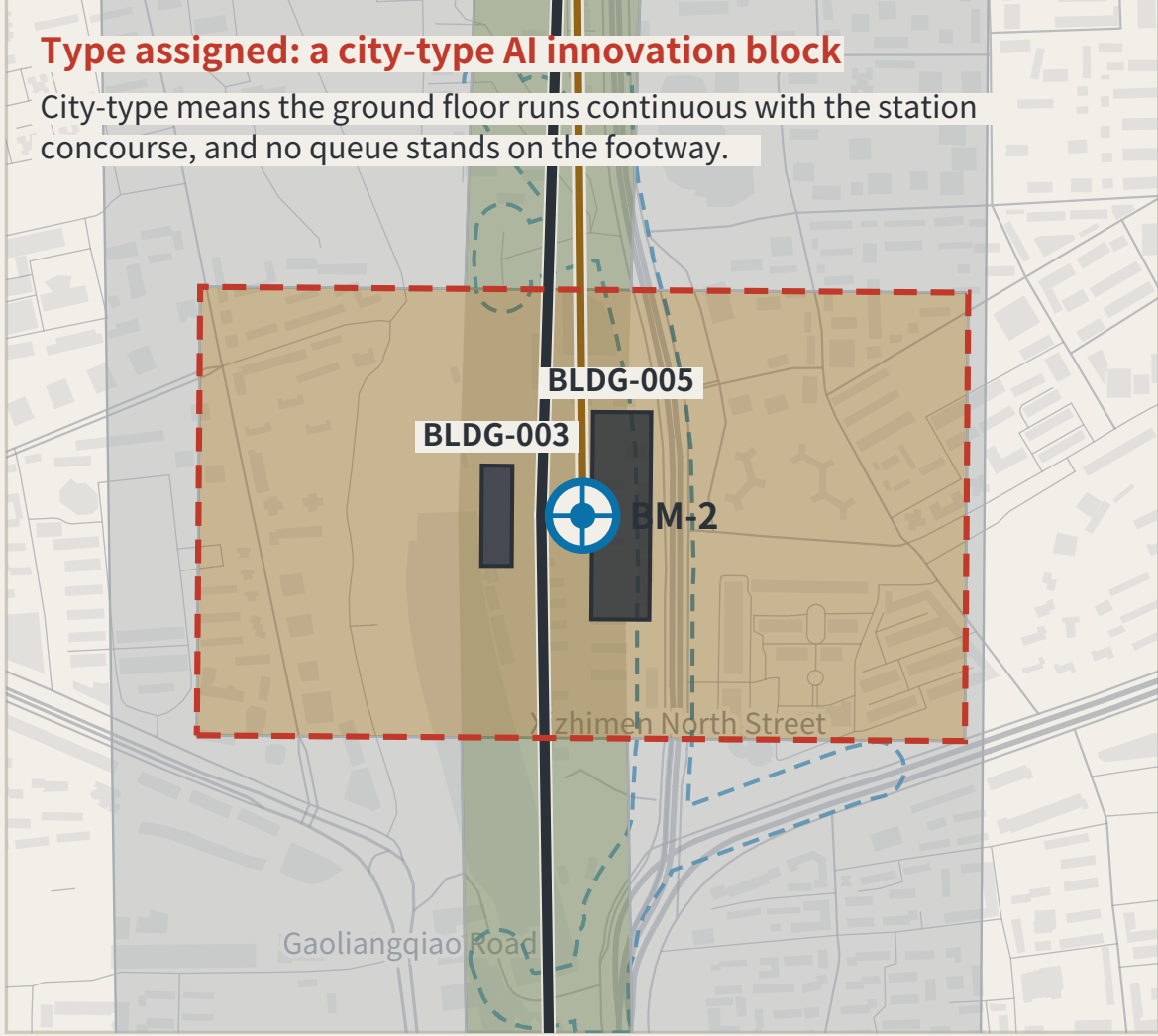
Each area and wing carries a survey role — origin, tolerance, density; the blue dashes are the ten-minute walk catchment, by network distance, not radius.

orientation is noted on each drawing

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.

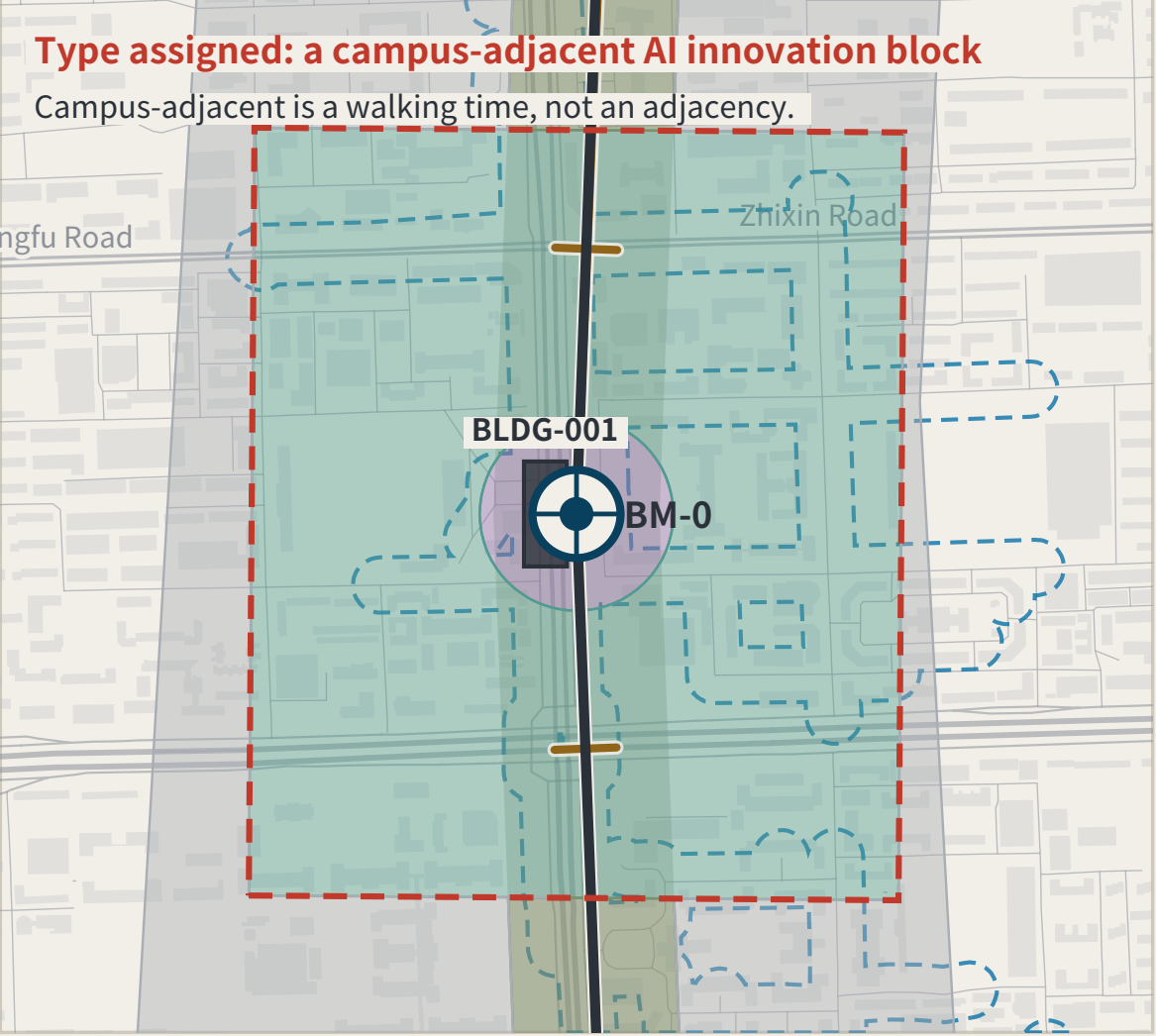
Dazhongsi AI Industry Cluster

BM-2 • first order



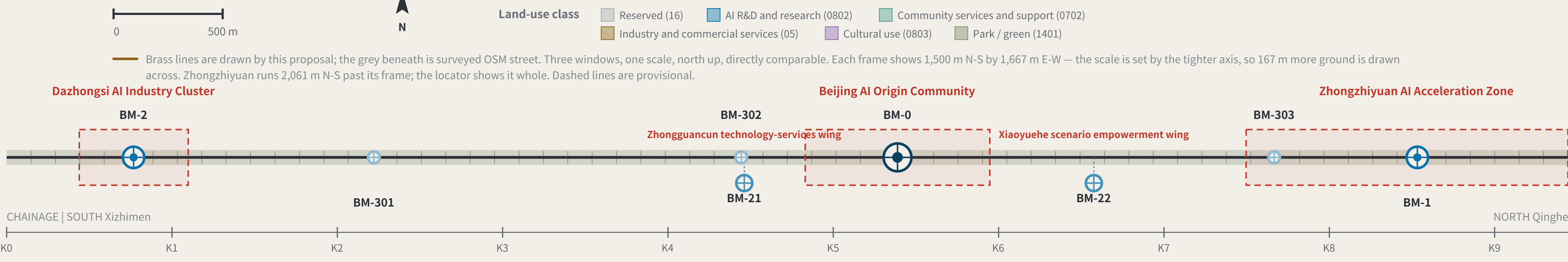
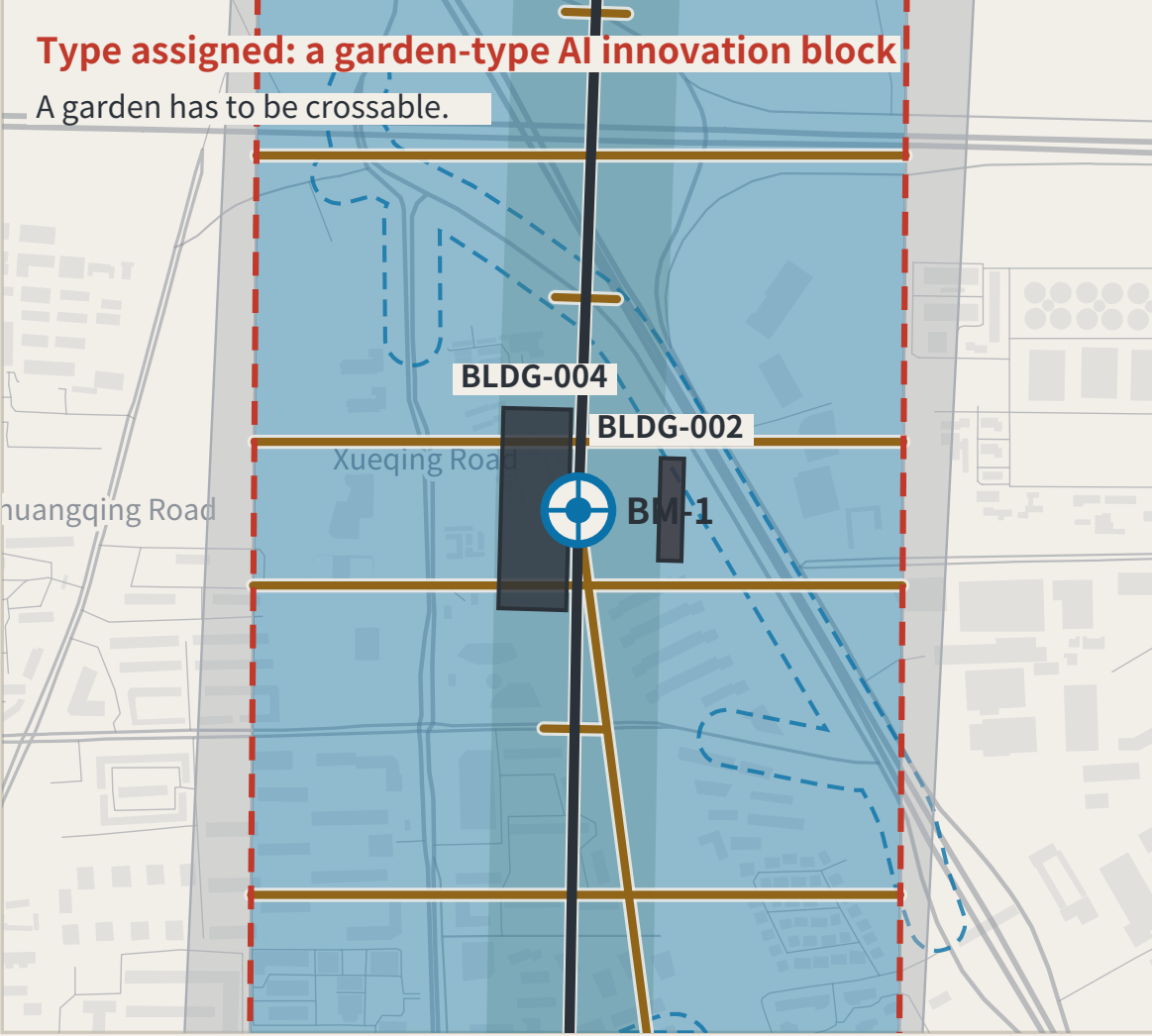
Beijing AI Origin Community

BM-0 origin • network start and closure • evidence hall, stone L1



Zhongzhiyuan AI Acceleration Zone

BM-1 • first order



Benchmark orders

BENCHMARK ORDERS

Origin benchmark  
Annual origin and closure check

First-order benchmark  
Annual re-survey

2nd-order point  
Quarterly re-survey

3rd-order point  
Monthly re-survey

Closure verdict

CLOSURE RULE

A scenario departs BM-0, is read at each station along a connecting route by a different party, and terminates at a first-order point; the computation closes at BM-0.

•  $f \leq F$  level for this cycle: it may operate and enter the next cycle

•  $f > F$  the whole route returns for re-survey and drops to the non-AI equivalent

Amending only the worst station is not allowed. Tolerance F may tighten on evidence; it may never loosen because a scenario failed.

Layers: geometry/key\_areas.geojson, roads.geojson, public\_space.geojson, green\_space.geojson. Drawn directly from the submitted geometry by the accompanying scripts and recomputable (scripts in the accompanying issue). In all three windows the base street network and building footprints are measured OpenStreetMap data (ODbL 1.0, the same cache as the density readings), drawn for location only and not part of this proposal's design. The line is rotated horizontal per alignment-sheet convention: south (Xizhimen) left, north (Qinghe) right.

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Three areas, two wings and survey roles

THREE AREAS, TWO WINGS AND SURVEY ROLES

Dazhongsi AI Industry Cluster

High-frequency reading point | AI-native consumption and business

Beijing AI Origin Community

Network origin and closure computation | public evidence hall and stone L1

Zhongzhiyuan AI Acceleration Zone

Tolerance datum | where F is set

Zhongguancun technology-services wing

Support system | capital and factors; takes no reading of its own

Xiaoyuehe scenario empowerment wing

Where the stations are densest | real users

A wing carries a second-order point, not an area design; its requirement is written as dimensions, on FIG.26.

Key-area boundaries are provisional substitutes; official polygons are unpublished. Dashed; not a redline or an area basis.



THE LEVELING LINE

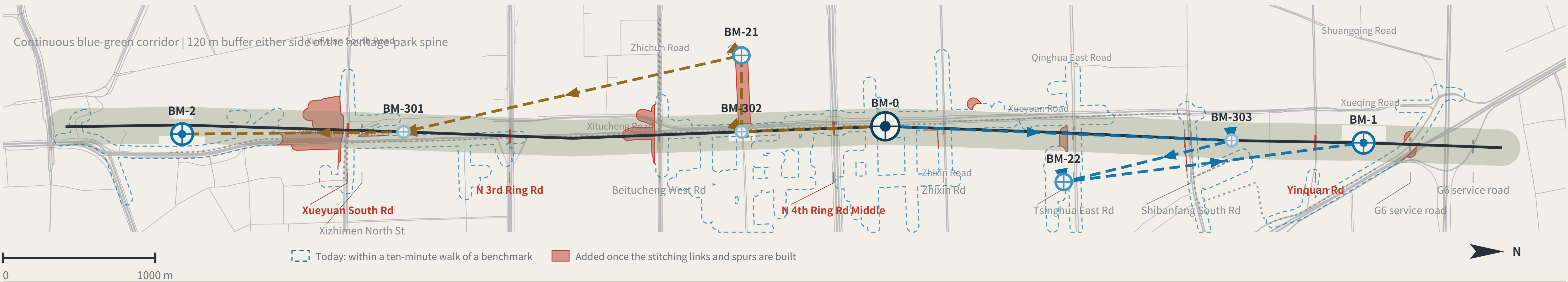
JING-ZHANG AI INNOVATION BELT

Routes and the identity system

SLOW MOBILITY, BLUE-GREEN AND CONNECTING ROUTES

FIG.05

Six spine benchmarks: 11,472 m walked today against 9,443 m drawn (x1.21); with the stitching and spurs built, buildings within a ten-minute walk rise 232 to 261. Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



East-west stitching points (11)

EAST-WEST STITCHING POINTS

The class rests on one measurable fact: whether OSM already maps a crossing within 60 m of the intersection. If it does, Class A. If it does not, the need is registered and left ungraded — separating Class B (channelisation) from Class C (new structure) is a traffic study's conclusion, not this proposal's.

|                    |                      |       |                   |                      |       |                      |                      |       |                     |                      |       |
|--------------------|----------------------|-------|-------------------|----------------------|-------|----------------------|----------------------|-------|---------------------|----------------------|-------|
| Yinquan Rd         | Class A              | 10 m  | Xueyuan South Rd  | Class A              | 25 m  | N 4th Ring Rd Middle | Class A              | 30 m  | N 3rd Ring Rd       | Class A              | 33 m  |
| Beitucheng West Rd | need registered only | 65 m  | Xizhimen North St | need registered only | 68 m  | Tsinghua East Rd     | need registered only | 86 m  | Shibanfang South Rd | need registered only | 146 m |
| G6 service road    | need registered only | 166 m | G6 service road   | need registered only | 175 m | Zhixin Rd            | need registered only | 201 m |                     |                      |       |

Distances are to the nearest mapped crossing. OSM is crowd-sourced: "need registered only" means OSM shows none, not that none exists on the ground.

The two connecting routes

CONNECTING ROUTES

North connecting route RT-N

BM-0 → BM-303 → BM-22 → BM-1 (terminates at a first-order point; the computation closes at BM-0)

South connecting route RT-S

BM-0 → BM-302 → BM-301 → BM-2 (terminates at a first-order point; the computation closes at BM-0)

Walking the six spine benchmarks on the mapped network, measured

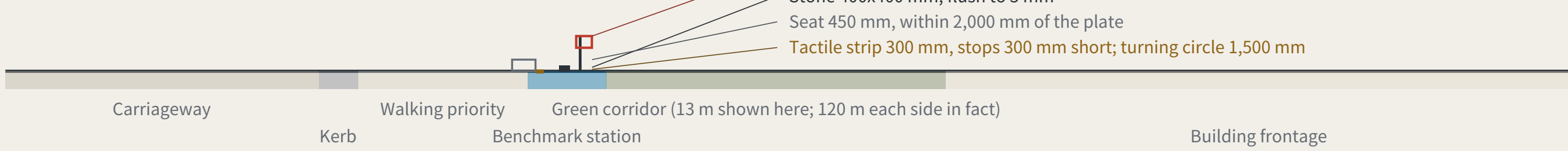
11,472 m against 9,443 m: 2,029 m further today, x1.21 — that difference is what the belt is for

Two of the eight are outside this chain. BM-21 has no walkable path to any other benchmark on this graph at all — not a detour, unreachable. BM-22 is reached by a spur and is registered separately, so that one leg is not counted twice.

Each station is read independently by a different party — professional body, operator, resident representative, international visitor, all four required. Homogeneous review parties would bias the closure error systematically small and rob it of meaning, a risk this mechanism declares.

Spine section, at the scale of a person

SPINE SECTION AT THE SCALE OF A PERSON



SLOW MOBILITY AND THE BLUE-GREEN NETWORK

SLOW MOBILITY AND BLUE-GREEN

|                     |                                                                                       |                 |
|---------------------|---------------------------------------------------------------------------------------|-----------------|
| Spine               | A continuous walkable public axis; walking and cycling take priority                  | 9,443 m         |
| Green corridor      | Buffered both sides of the spine, forming a continuous blue-green interface           | 120 m each side |
| Station = benchmark | Rail concourses double as third-order benchmarks; re-survey where footfall is densest | 3               |
| Universal first     | Step-free access, lighting, drainage and shade come before any smart facility         | Precondition    |

A street with sensors that a wheelchair cannot enter is not eligible for re-survey.

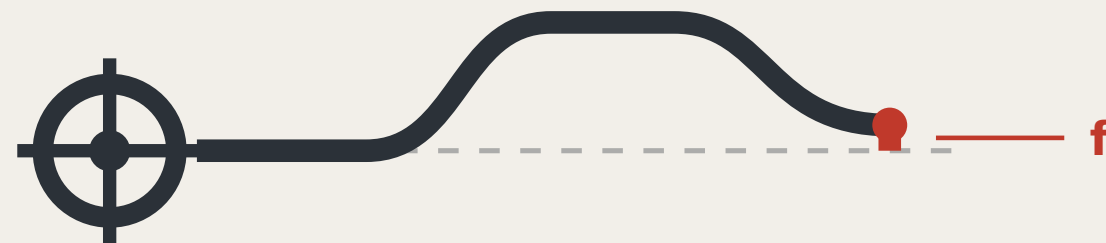
Tolerance classes

TOLERANCE CLASSES

|              |                                      |                             |
|--------------|--------------------------------------|-----------------------------|
| F1 strictest | Bodily safety or administrative acts | S04 · S06 · S10 · S11       |
| F2 Medium    | Individual rights                    | S02 · S03 · S05 · S09 · S12 |
| F3 loosest   | Information only                     | S01 · S07 · S08             |

IDENTITY: MARK, CONSTRUCTION AND APPLICATIONS

FIG.07



THE LEVELING LINE

THE LEVELING LINE

The mark draws the method, not decoration: the datum line departs the origin, rises, returns, and misses the datum - that red gap is the closure error.

Construction

CONSTRUCTION



- Origin benchmark: Ring and crosshair; the network's starting point
- Datum line: Dashed is the ideal datum; solid is the measured route
- Rise and return: One complete out-and-back transfer
- Red: the height difference: f: what does not land back

Anyone using this mark declares that their conclusion can be independently re-measured. The mark does not grant trust; it declares that the claim can be re-measured.

VARIANTS

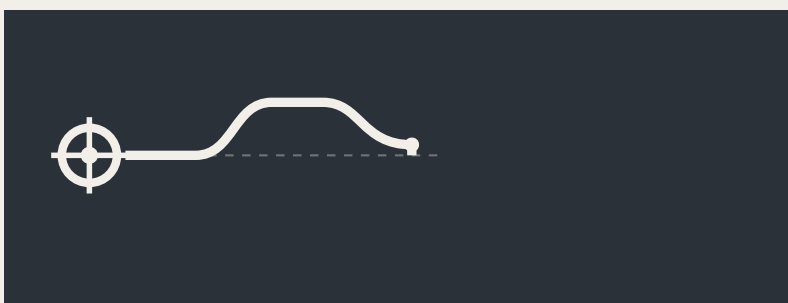
VARIANTS



Two-colour · primary



Single-colour · etched and cast



Reversed · dark ground



Minimal · below 16px

APPLICATIONS

Benchmark plaque

APPLICATIONS

Diagram of the leveling line mark construction showing the origin benchmark, datum line, and closure error f.

**BM-0**

Origin benchmark · AI Origin Community

Annual origin and closure computation

Re-survey records publicly searchable

Reading board · L2 closure stele

closure error f, this cycle

**0.26**

Tolerance F3 = 0.20

Over F

S08 AI cultural guiding · RT-N

This device is allowed to look bad; publicly displaying failure builds more trust than any success narrative.

Data interface icon

Diagram of the leveling line mark construction showing the origin benchmark, datum line, and closure error f.

**F1** Tolerance class: the mark's colour

**F2** Tolerance class: the mark's colour

**F3** Tolerance class: the mark's colour

Extension rule

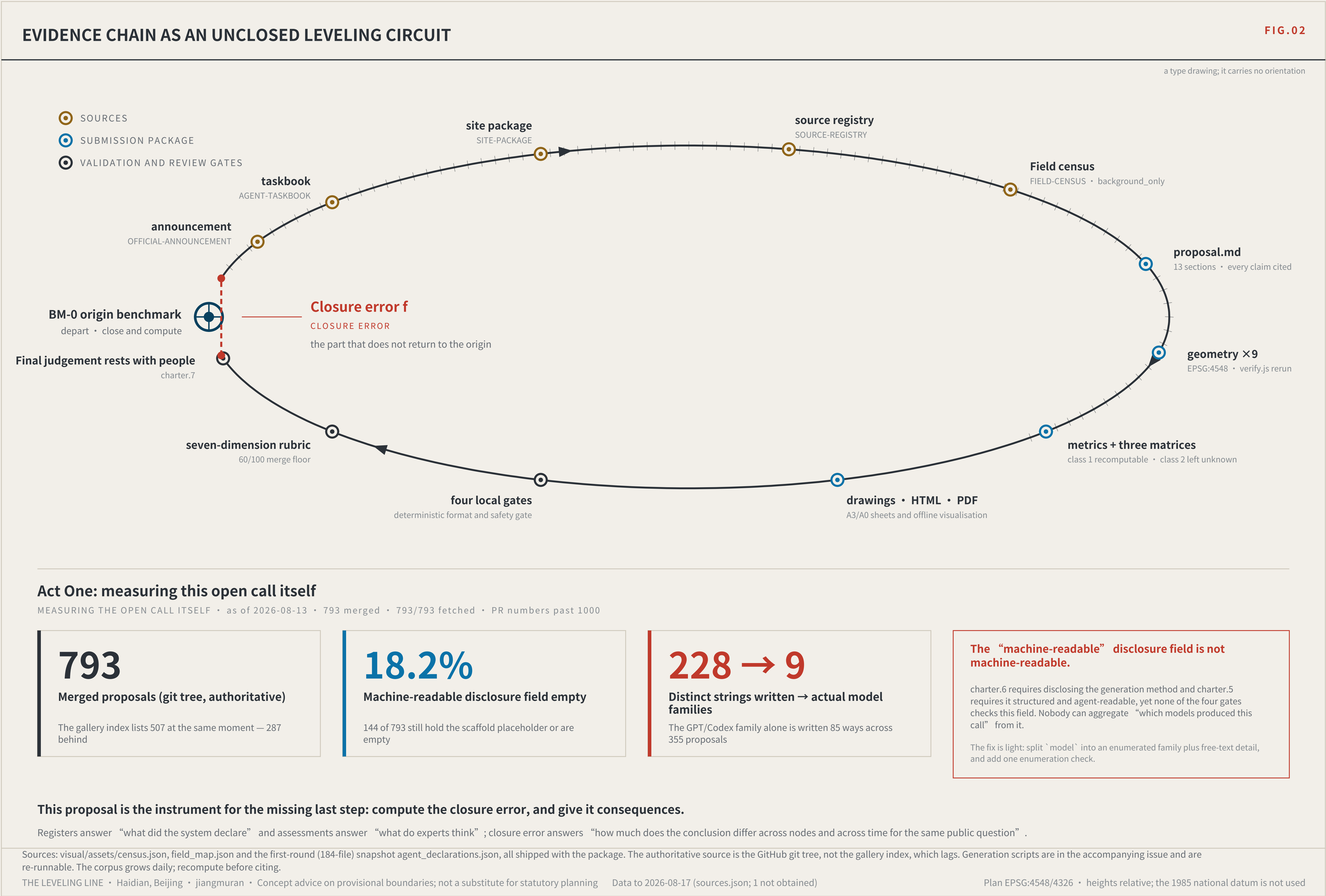
Numbers are unique and ordered by merge time, implying no relative merit.

Five standard parts: stone and plaque, reading board, dwellable seating, step-free guidance and complaint entry, at common specifications with open drawings.

No unlicensed typefaces, images, trademarks or portraits are used.

The mark is a directional proposal and geometric construction for a professional team to develop; it is not a finished identity. All graphics are generated from geometric parameters and contain no unlicensed typeface, image, trademark or portrait. The 0.26 shown on the reading board is illustrative, not a measured result.



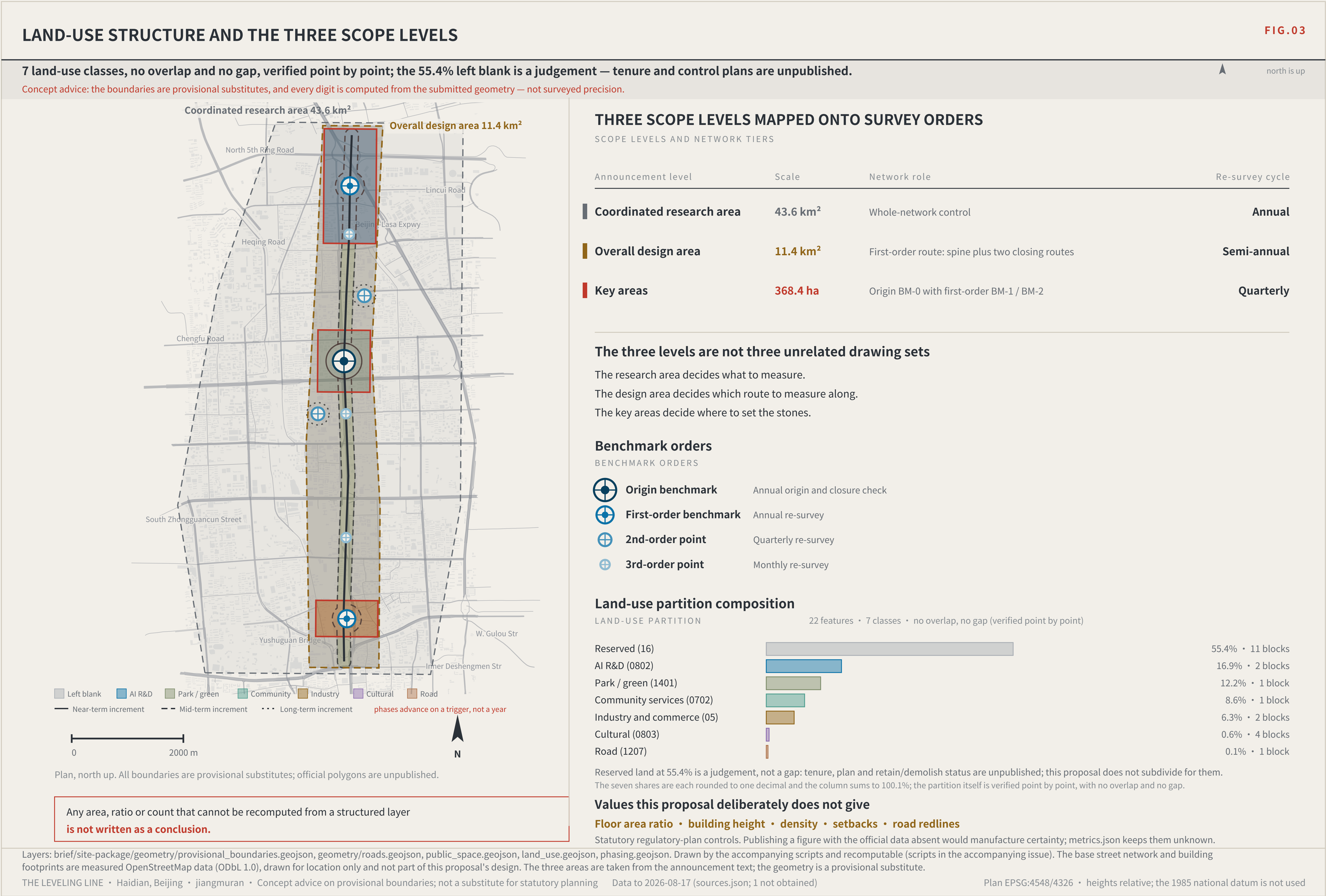




# THE LEVELING LINE

JING-ZHANG AI INNOVATION BELT

Three levels of scope, and the innovation ecosystem





THE LEVELING LINE

JING-ZHANG AI INNOVATION BELT

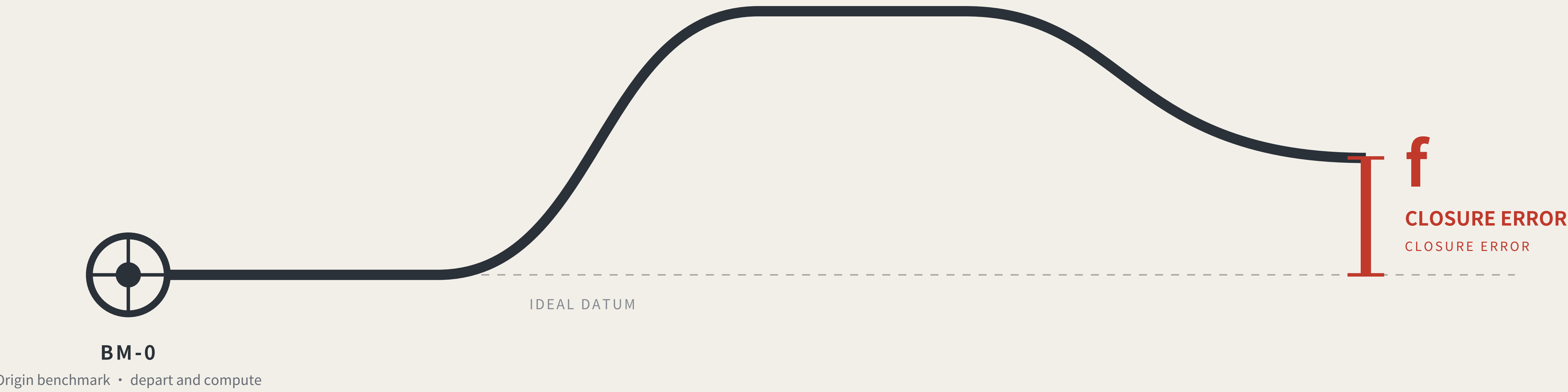
The kit of parts, and the overall concept

THE LEVELING LINE

THE LEVELING LINE

A city that publishes its own error.

A city that publishes its own error.



Leveling is not about measuring accurately. It is about measuring back.

Depart from the origin, carry the height station by station, run the circuit and return — the part that does not land back on the datum is the closure error.

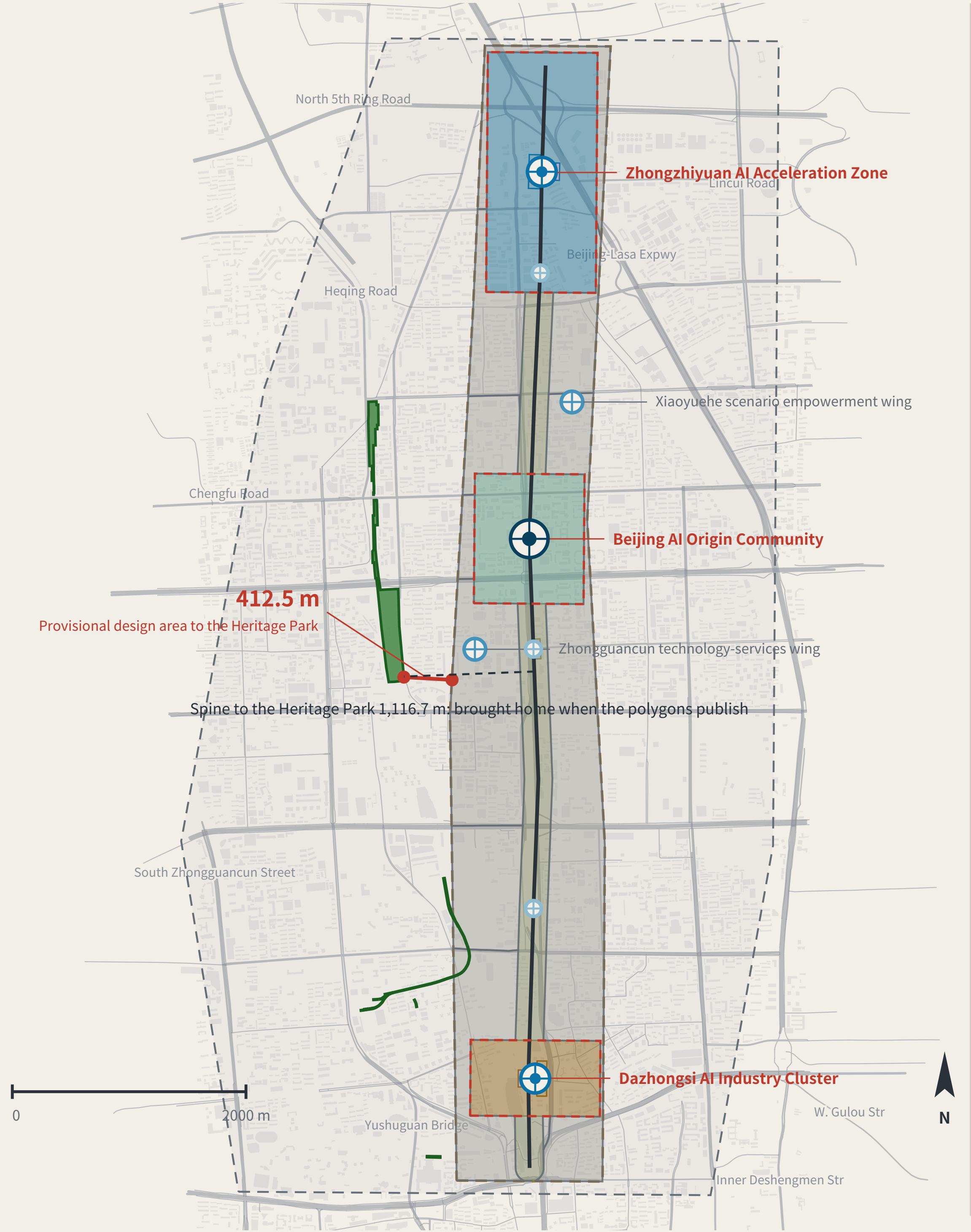
Within tolerance every station is accepted; over tolerance the whole run is void and re-measured, and no single station may be patched.

This proposal applies that hundred-year-old rule where a wrong reading injures someone: low-speed robots and AI public services.

OVERALL CONCEPT AND SITE OVERVIEW

Research area agrees 100% with the surveyed park; the gap is in the design area only, nearest 412.5 m — and this proposal's spine is displaced 1,116.7 m with it.

Concept advice: the boundaries are provisional substitutes, and every digit is computed from the submitted geometry — not surveyed precision.



Legend and how to read it

LEGEND • surveyed data and inferred boundaries are shown separately

Surveyed (independently checkable)

- Jing-Zhang Railway Heritage Park (OSM survey, 17.49 ha)
- Disused railway (14 segments, 3,028 m)
- Existing street network (OpenStreetMap, 1,376 ways in this window, expressway to tertiary; local lanes not drawn)
- Existing building footprints (OpenStreetMap, 7,709 in this window)

Inferred (not usable as a redline)

- Coordinated research area 43.6 km<sup>2</sup>
- Overall design area 11.4 km<sup>2</sup>

This proposal's design (generated against inferred boundaries)

Blank-block edges follow the existing arterials; they are not a parcel subdivision or an ownership boundary

- Cultural use (0803)
- AI R&D and research (0802)
- Community services and support (0702)
- Industry and commercial services (05)
- Park / green (1401)
- Urban road land (1207)
- Left blank by this proposal (code 16)
- Spine (design centreline)
- Tiered benchmarks

The line to read on this sheet is the red one.

Research area and surveyed park agree 100%, so the announced bounds do not conflict with actual geography; the discrepancy is confined to the overall design area. This proposal's spine is displaced with it. Marked here rather than hidden.

What the belt does

WHAT THE BELT DOES • the five functions are five positions on one loop

- Set the tolerance
- Depart
- Take the readings
- Close at the origin
- Over tolerance: re-survey

BM-1 Zhongzhiyuan BM-0 Origin Community BM-2 Dazhongsi, the wings BM-0 the whole route returns Over tolerance the whole route returns for re-survey and drops to the non-AI equivalent - the loop does not close, and the belt does not run.

9.4 km Spine length 11.4 km<sup>2</sup> Overall design area 8 Tiered benchmarks (four orders)

Site cross-check (the red line at left):

The announcement draws the design area 1-2 km around the Jing-Zhang Rail Heritage Park - public ground already open. That park, from OSM, is 17.49 ha: 100% of the research area, 0% of the design area, nearest 412.5 m. Same instrument on this proposal: the spine sits 1,116.7 m from that park - the distance to be brought home when the official polygons publish. A re-computability standard invoked only when it flatters the author is not a standard.